

Lab 3 SE

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Justification for Architecture choice:

I chose microservices architecture for the Personal Subscription and Recurrence auditor.

Analysis:

When evaluating other patterns for this system, both Layered and Client-Server architectures present limitations. Layered Architecture is organized in horizontal layers (such as Presentation, Business, and Data). While this provides a clear separation of concerns and is very common in traditional full-stack web applications, it suffers from tight coupling between layers and makes it difficult to scale individual components. For our auditor, this means the intensive transaction analysis couldn't be scaled independently of the user interface. Client-Server Architecture relies on a centralized server with multiple clients. Although this offers centralized control and easy data consistency, it creates a single point of failure and a scalability bottleneck, meaning an overwhelming influx of imported CSV/OFX data could bottleneck the entire server and crash the application for all users.

. **Decoupled Analytics (Alignment with FR001 & FR002):** The system must allow users to import transaction data via CSV or OFX formats and subsequently analyze this data to identify recurring billing patterns. A microservices approach allows the transaction analysis engine to operate independently. This separation makes it possible to deploy specialized solutions within the analysis component-such as utilizing lightweight language models fine-tuned with QLoRA to accurately extract and parse transaction descriptions, amounts, and dates-without bloating the main application.

. **Dedicated Asynchronous Operations (Alignment with FR004):** The system is required to identify upcoming subscription renewal dates and issue advance alerts to the Individual User. By utilizing an independent notification microservice, the system can continuously run scheduled checks in the background to generate these alerts before the renewal date, completely separate from the user's active session.

Security Advantage: The non-functional requirements explicitly state that the system must protect sensitive user data by securely processing transaction imports without storing raw banking credentials in cleartext. Microservices provide a distinct security advantage by allowing the data ingestion and database management components to be isolated within a highly restricted private network. This strict segregation ensures that even if the public-facing application is compromised, the sensitive financial data and credential stores remain protected.

Performance Benefit: The system is required to process and analyze large transaction files within an acceptable response time. Identifying recurring subscriptions requires grouping transactions and checking their frequency and consistency. Because this transaction analysis is computationally intensive, isolating it into a standalone service allows it to scale dynamically under heavy loads. This ensures that processing a large transaction file does not exceed the defined performance targets or

cause excessive delays for the user while they are viewing their estimated monthly subscription expenditure.

Drawbacks: Microservices architecture introduces additional complexity, network latency and data consistency challenges.

Requirement ID	Requirement Description	Mapped Components	Rationale
FR001	The system shall analyse imported transactions, identify recurring billing patterns as subscriptions.	Transaction Analyser	This component contains the core logic for identifying grouping frequency and consistency.
FR002	The system shall allow the Individual User to import transaction data from formats such as CSV or OFX.	Auditor Web UI, Bank File Importer	The UI receives the file from the user, and the Importer parses the specific CSV/OFX formats.
FR003	The system shall calculate, display the user's estimated monthly subscription expenditure based on recurring subscriptions.	Transaction Analyzer, Auditor Web UI	The Analyzer calculates the expenditure, and the Web UI handles the display to the user.
FR004	The system shall identify subscription renewal dates and alert the Individual User before renewal date.	Notification Engine	This background component handles the asynchronous task of dispatching the advance notifications.
FR005	The system provides cancellation guidance for subscriptions.	Subscription Manager, Auditor Web UI	The Manager retrieves the specific instruction sets for the selected subscription, then rendered on the UI.
NFR001	The system shall process and analyse imported transaction data within an acceptable response time, even when handling a large transaction file.	Transaction Analyzer, Bank File Importer	These components must be optimized to handle large computational loads without bottlenecking the system.
NFR002	The system shall securely process financial transaction imports without storing raw banking credentials in cleartext.	Secure Data Repository, Bank File Importer	The Importer safely handles the initial data, and the Repository enforces the encryption and strict storage protocols.

